

Project Report: Titanic Survival Prediction

Author: Bassam Kasar

Internship: CodSoft Data Science Intern

Date: May 2026

1. Introduction

The Titanic Survival Prediction project is a classic data science challenge that involves predicting whether a passenger survived the 1912 shipwreck based on personal and demographic data. This project serves as a foundational exercise in **binary classification**, data cleaning, and exploratory data analysis.

2. Problem Statement

The goal is to build a machine learning model that uses passenger data (such as age, gender, and socio-economic status) to predict the survival status of individuals. The target variable is Survived, where **1** indicates survival and **0** indicates deceased.

3. Dataset Description

The project utilizes the Titanic-Dataset.csv file. The primary features analyzed include:

- **Pclass (Passenger Class):** A proxy for socio-economic status (1st, 2nd, 3rd).
- **Sex:** Gender of the passenger.
- **Age:** Age in years (included many missing values).
- **SibSp / Parch:** Number of family members (siblings, spouses, parents, children) aboard.
- **Fare:** Passenger fare (ticket price).
- **Embarked:** Port of embarkation (C = Cherbourg; Q = Queenstown; S = Southampton).

4. Methodology

A. Data Preprocessing

To ensure model accuracy, the raw data underwent several cleaning steps:

- **Feature Selection:** Dropped PassengerId, Name, Ticket, and Cabin as they contained unique strings or too many null values to be statistically significant.
- **Imputation:**
 - Missing **Age** values were replaced with the **median** age.
 - Missing **Embarked** values were filled with the **mode** (most frequent port).
- **Categorical Encoding:** Used LabelEncoder to convert 'Sex' and 'Embarked' text into numerical format so the algorithm could process the data.

B. Model Selection

A **Random Forest Classifier** was selected for this task.

- **Why Random Forest?** It is an ensemble learning method that creates multiple decision trees and merges them to get a more stable and accurate prediction. It handles both categorical and numerical data effectively and is less prone to overfitting than a single decision tree.

C. Training & Testing

The dataset was split using an **80/20 ratio**:

- **80% (Training):** Used to "teach" the model the relationship between features and survival.
- **20% (Testing):** Used to evaluate the model's performance on unseen data.

5. Results and Analysis

Model Performance

- **Accuracy:** The model achieved an accuracy of approximately **81.56%**.
- **Precision/Recall:** The model showed high precision in identifying non-survivors, with a slightly lower recall for survivors, indicating a conservative prediction strategy.

Feature Importance

The model identified the following features as the most influential in determining survival:

1. **Sex:** The highest indicator (due to "women and children first" protocols).
2. **Fare & Pclass:** High-income passengers and those in 1st class had significantly higher survival rates.
3. **Age:** Younger passengers generally had a higher likelihood of survival.

6. Conclusion

The project successfully demonstrates how machine learning can be used to analyze historical data and find patterns that lead to predictive insights. By handling missing data and selecting an appropriate ensemble model, we were able to predict survival with over 80% accuracy.